

R Code for session_duration_ttest

Sidney Reynaud

2026-07-20

```
#####  
# Step 1: Load required packages  
#####  
library(dplyr)  
  
## Warning: package 'dplyr' was built under R version 4.3.3  
  
##  
## Attaching package: 'dplyr'  
  
## The following objects are masked from 'package:stats':  
##  
## filter, lag  
  
## The following objects are masked from 'package:base':  
##  
## intersect, setdiff, setequal, union  
  
#####  
# Step 2: Import synthetic dataset  
#####  
data <- read.csv("practical-application/synthetic_data.csv")  
  
#####  
# Step 3: Run one-tailed t-test  
# Hypothesis: Completers (Conversion = 1) have  
# longer session durations than abandoners (0)  
#####  
session_test <- t.test(  
  TimeSpent ~ Conversion,  
  data = data,  
  alternative = "greater" # one-tailed test  
)  
  
#####  
# Step 4: Save t-test results to text file  
#####  
capture.output(  
  session_test,  
  file = "outputs-test_results/session_duration_ttest.txt"  
)
```

```
#####  
# Step 5: (Optional) Print to console  
#####  
print(session_test)
```

```
##  
## Welch Two Sample t-test  
##  
## data: TimeSpent by Conversion  
## t = 0.95033, df = 85.508, p-value = 0.1723  
## alternative hypothesis: true difference in means between group 0 and group 1 is greater than 0  
## 95 percent confidence interval:  
## -0.640099 Inf  
## sample estimates:  
## mean in group 0 mean in group 1  
## 7.276285 6.422560
```